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Received: 17 May 2026

Accepted: 22 July 2026

Published online: 27 July 2026

Cite this article as: tao W., haijie Q., yanli X. *et al.* Single-center clinical analysis of anomalous aortic origin of a coronary artery (AAOCA) in children: a multimodal imaging approach to risk stratification. *J Cardiothorac Surg* (2026). <https://doi.org/10.1186/s13019-026-04667-9>

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Single-Center Clinical Analysis of Anomalous Aortic Origin of a Coronary Artery (AAOCA) in Children: A Multimodal Imaging Approach to Risk Stratification

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Abstract

Objective This study aimed to analyze the clinical characteristics of pediatric patients with Anomalous Aortic Origin of a Coronary Artery (AAOCA), explore the application of multimodal imaging in risk stratification, and establish individualized diagnosis and treatment protocols for this population.

Methods A retrospective observational study was performed on children diagnosed with AAOCA at Wuhan Children's Hospital between January 2015 and April 2026. Clinical manifestations, multimodal imaging findings, therapeutic strategies, and long-term outcomes were comprehensively analyzed.

Results A total of 58 AAOCA patients confirmed by coronary computed tomography angiography (CCTA) were enrolled. Initial transthoracic echocardiography (TTE) yielded false-negative results in 24 patients (41.4%). Twenty-seven patients underwent single-photon emission computed tomography myocardial perfusion imaging (SPECT-MPI, MPS), among whom 6 had normal perfusion and 21 showed perfusion defects. Seven patients with positive MPS findings received surgical correction, including 2 patients with L-AAOCA (1 coronary angioplasty and 1 coronary unroofing) and 5 patients with R-AAOCA; all treated with coronary unroofing. No patients with negative MPS results underwent surgery, indicating that negative MPS findings perhaps provide limited valuable evidence to avoid unnecessary coronary revascularization. Two patients developed massive postoperative pericardial effusion and recovered after pericardiotomy and drainage; the remaining 5 surgical patients experienced no perioperative complications. All surgical patients were discharged after full recovery and received 3 months of aspirin antiplatelet therapy. Non-surgical patients were instructed to restrict strenuous physical activity and followed up for 3 months to 3 years. Follow-up CCTA or TTE revealed normal coronary ostia without stenosis in all patients.

Conclusion Clinical manifestations of pediatric AAOCA vary widely across age groups, ranging from asymptomatic presentation to sudden cardiac arrest

(SCA), leading to marked discrepancies in risk evaluation and therapeutic decision-making. Multimodal imaging including TTE, CCTA and SPECT-MPI facilitates risk stratification and guides personalized clinical management of children with AAOCA.

Keywords anomalous aortic origin of a coronary artery(AAOCA); multimodal imaging; coronary unroofing; single-photon emission computed tomography myocardial perfusion imaging;SPECT-MPI

1 Introduction

Anomalous Aortic Origin of a Coronary Artery (AAOCA) is a rare congenital heart defect defined as the aberrant origin of a coronary artery from an inappropriate aortic sinus or ascending aortic segment. Most AAOCA cases are identified incidentally without obvious clinical symptoms. Nevertheless, approximately 20% of affected individuals develop coronary insufficiency, which may progress to myocardial infarction, refractory arrhythmia, sudden cardiac arrest (SCA) and sudden cardiac death (SCD). AAOCA is recognized as the second most common cause of exercise-related SCD in young athletes.

This single-center retrospective study systematically analyzed clinical and multimodal coronary imaging data of pediatric AAOCA patients to implement standardized risk stratification and individualized treatment strategies, with the ultimate goal of minimizing unwarranted exercise restrictions and redundant surgical interventions.

2 Materials and Methods

2.1 Study Population

This retrospective study enrolled 58 children with CCTA-confirmed AAOCA treated at Wuhan Children's Hospital from January 2015 to April 2026. The research protocol was approved by the Institutional Ethics Committee of Wuhan Children's Hospital (Approval No: 2026R050-E01).

2.2 Data Collection and Inclusion Criteria for SPECT-MPI

Clinical data were extracted from the hospital electronic medical record system, including demographic data (age, sex, body weight), clinical symptoms, myocardial injury biomarkers (creatinine kinase-MB [CK-MB], cardiac troponin I [TnI]), and auxiliary examination results (12-lead electrocardiography [ECG], TTE, CCTA, SPECT-MPI). Patients complicated with cardiomyopathy, complex congenital heart disease, other congenital coronary malformations or acquired coronary artery lesions were excluded.

SPECT-MPI was indicated in patients who met at least one of the following criteria:

- (1) presence of ischemia-related symptoms (e.g., chest pain, syncope, or exertional dyspnea);
- (2) abnormal electrocardiographic findings suggestive of ischemia;
- (3) elevated cardiac biomarkers indicating myocardial injury; or
- (4) high-risk anatomical features on coronary CTA, such as an intramural course, slit-like orifice, or acute takeoff angle of the anomalous coronary artery.

2.3 Electrocardiography

All participants underwent standard 12-lead surface ECG. Tracings were analyzed for abnormal Q waves and ST-T segment alterations suggestive of myocardial ischemia.

2.4 Transthoracic Echocardiography

TTE examinations were completed using a Philips color Doppler ultrasound system. Comprehensive assessments of cardiac morphology, valvular function and coronary artery anatomy were performed for both preoperative evaluation and postoperative follow-up.

2.5 Coronary Computed Tomography Angiography

CCTA was acquired on a Siemens dual-source CT scanner with an effective radiation dose controlled below 2 mSv. Non-ionic contrast medium (300–370 mgI/mL) was administered intravenously at a dose of 1.5–2.0 mL/kg. Thin-slice reconstruction (0.5 mm) was performed post-scanning, followed by volume rendering (VR), maximum intensity projection (MIP) and curved planar reformation (CPR) to delineate coronary artery anatomy and intracardiac structures.

2.6 SPECT-MPI Protocol and Image Analysis

^{99m}Tc-MIBI was used as the radiotracer for pediatric myocardial perfusion imaging. Resting radiotracer dosage was standardized by body weight (0.1–0.15 mCi/kg; total range: 2.5–5.0 mCi [92.5–185 MBq]) via intravenous injection, with scanning performed on a SPECT/CT system. High-level iterative reconstruction was applied for pediatric image post-processing. Gated images were semi-quantitatively analyzed using Myovation Evolution software. Two experienced nuclear medicine physicians independently reviewed all images in a double-blinded manner.

Image interpretation followed the AHA 17-segment myocardial model with a 5-point scoring system:

0 = normal radiotracer uptake;

1 = mild reduction;

2 = moderate reduction;

3 = severe reduction;

4 = complete absence of uptake.

A score ≥ 1 in two consecutive slices with corresponding defects on orthogonal views was defined as a positive perfusion defect.

2.7 Treatment Regimens

Surgical interventions included coronary unroofing and coronary angioplasty. Conservative management consisted of oral metoprolol (β -blocker) and strict restriction of strenuous physical exertion to avoid exercise-induced ischemia.

2.8 Follow-Up Protocol

All patients received regular follow-up ranging from 3 months to 3 years. Primary observational endpoints included syncope, arrhythmia and impaired cardiac function. Routine follow-up examinations contained ECG, TTE and treadmill exercise testing; repeat CCTA was arranged when clinically indicated.

2.9 Statistical Analysis

Statistical analyses were conducted using SPSS 26.0 (IBM Corp., Armonk, NY, USA). Categorical variables were presented as case numbers (percentages). Intergroup comparisons were performed using Pearson's χ^2 test.

3 Results

Fifty-eight CCTA-confirmed AAOCA children were included, consisting of 54 R-AAOCA and 4 L-AAOCA patients. The cohort included 39 males and 19 females, aged 3 months to 17 years with a mean age of 10.23 ± 0.42 years. Patients were stratified into four age subgroups: infants (<3 years), preschool children (3–6 years), school-age children (6–12 years), and adolescents (12–18 years) (table 1). Most infants and preschoolers were incidentally diagnosed during hospital visits for fever, vomiting and other unrelated complaints, with non-statistically significant fluctuations in myocardial biomarkers. In contrast, chest tightness and chest pain were the predominant initial symptoms in school-age and adolescent patients, and 2 patients presented with syncope. No SCD events occurred during follow-up, and all symptomatic school-age and adolescent patients had normal myocardial biomarker levels. Initial TTE failed to identify AAOCA in 24 patients (41.4%) (table 2).

Table 1 Clinical characteristics of cases

Variable, n=58	palpitation	Chest pain and tightness	syncope	unrelated complaints*
male, n(%)	18(1.72)	16(27.6)	3(5.2)	16(27.6)
female, n(%)	18(1.72)	12(20.7)	3(5.2)	6(10.3)
<3 years, n(%)	0(0)	0(0)	0(0)	6(10.3)
3-6 years, n(%)	0(0)	0(0)	0(0)	2(3.4)
6-12 years, n(%)	0(0)	19(32.8)	4(0)	11(19.0)
12-18 years, n(%)	2(3.4)	9(15.5)	2(0)	3(5.2)

*unrelated complaints:fever, vomiting,etc

Table 2 Clinical characteristics of AAOCA

Variable	sort	ARCA (n=54,%)	ALCA (n=4,%)
sex	male	37 (68.5)	2 (50.0)
	female	17 (31.5)	2 (50.0)
Clinical symptom ,	thoracodynia	26 (48.1)	2 (50.0)
n(%)	palpitation	2 (3.7)	1 (25.0)
	syncope	5 (9.3)	1 (25.0)
	unrelated complaints*	22 (40.7)	0 (0.0)
electrocardiogram, (%)	normal	36 (66.7)	2 (50.0)
n(%)	arrhythmia	13 (24.1)	1 (25.0)
	Myocardial ischemia	5 (9.3)	1 (25.0)
Cardiac CTA	Completed/positive	54 (100.0)	4 (100.0)
MPS	Completed/positive	24 (44.4)	3 (75.0)

*unrelated complaints:fever, vomiting,etc

Twenty-seven patients completed SPECT-MPI: 6 patients with negative perfusion findings received no surgical intervention, while 21 patients with positive perfusion defects underwent evaluation for surgery, among whom 7 received coronary revascularization. Surgical cases included 2 L-AOOCA patients (1 coronary angioplasty, 1 coronary unroofing) and 5 R-AOOCA patients who all underwent coronary unroofing. The mean operative duration was 1.5 ± 0.35 hours. Two patients developed massive postoperative pericardial effusion and underwent emergency pericardiotomy and drainage; no other postoperative complications were recorded in the remaining 5 surgical patients (table 3). All surgical patients recovered fully and were discharged, with a 3-month postoperative aspirin antiplatelet course. Postoperative TTE was performed at 3-6 months in 2 patients, and CCTA at 1-3 years in 5 patients, which confirmed normal coronary ostial position without stenosis. Non-surgical patients were advised to limit physical activity and followed up for 3 months to 3 years.

Notably, the L-AOOCA subgroup only contained 4 patients, leading to limited statistical power for intergroup comparison, and cross-group causal inference was not feasible. Interpretations of the present findings should be cautious given the small single-center sample size; large-cohort multicenter studies are required to validate these results. Fisher's exact test was not applicable due to insufficient sample size in the L-AOOCA arm.

Table 3 Clinical data of children undergoing surgery

case	sex	Age	Action in chief	Myocardial marker	ECG	TTE	CTA	MPS (reduction area of myocardial color scale distribution, and percent of the total left ventricular area)	treatment
case1	Female	13 years	chest pain	normal	normal	normal	ALCA, interarterial	anterior wall near the apex of the left ventricle, 4%	angioplasty
case2	Male	11 years	chest tightness	normal	normal	normal	ARCA, proximal narrowing	apex of the heart, anterior parapical segment, middle part, 14%	unroofing
case3	Male	6 years	Postoperative dilatation of PS	normal	Sinus arrhythmia	ALCA	ALCA, proximal narrowing, interarterial	apex of the heart, 2%	unroofing
case4	Female	12 years	chest tightness	normal	T-wave change	ARCA	ARCA, intermural, proximal narrowing	in the apex of the heart, anterior wall and the middle part of the heart, 4%	unroofing
case5	Female	10 years	chest pain	normal	normal	ARCA	ARCA, intermural, proximal narrowing	in the anterior apical segment and the middle part of the left ventricle, 13%	unroofing
case6	Male	10 year	chest pain and tightness	normal	ST segment elevation	Right crown high origin	ARCA, intermural, proximal narrowing	in the proximal apical segment of the lower wall of the left ventricle, 1%	unroofing
case7 (Figure 1)	Male	9 years	chest tightness and pain	normal	Ectopic room rule, no ST change	normal	ARCA, intermural, proximal narrowing	in the anterior apical, anterior middle, inferior apical; 13%	unroofing

ECG, electrocardiogram; TTE, echocardiography; CTA, coronary computed tomography angiography; MPS, single photon emission tomography myocardial perfusion imaging; Myocardial markers: Troponin I, CK-MB

4 Discussion

AAOCA is a rare congenital coronary malformation characterized by anomalous coronary origin from an inappropriate aortic sinus or ascending aorta, classified into R-AAOCA, L-AAOCA and single coronary artery anomalies arising from the left or right aortic sinus^[1]. Clinical manifestations of pediatric AAOCA are non-specific. Most cases are detected incidentally on routine TTE during unrelated outpatient visits, while only a minority present with chest pain, palpitations or syncope^[2]. Conventional examinations including myocardial injury biomarkers, resting ECG and TTE lack diagnostic specificity for AAOCA, and TTE carries a considerable false-negative rate^[3]. Approximately 20% of AAOCA patients develop pathological coronary ischemia, which predisposes individuals to life-threatening complications such as arrhythmia, SCA and SCD^[4]. Domestic studies have previously reported six pediatric cases of SCA secondary to coronary malformations requiring extracorporeal membrane oxygenation (ECMO) support.

Early pathophysiological hypotheses postulated that strenuous exercise-induced dilation of the aorta and pulmonary artery compresses the interarterial coronary segment and triggers ischemia^[5]. However, modern myocardial perfusion imaging and mechanistic studies have revised this theory: exercise-related stretching of the aortic wall narrows the intramural coronary lumen and causes fixed obstruction secondary to slit-like ostia and proximal stenosis, alongside dynamic obstruction driven by acute coronary takeoff angles, intramural course and elliptical vessel morphology^[6]. Pulmonary artery compression rarely contributes to ischemia in the absence of pulmonary hypertension or pulmonary arterial dilation^[7]. This study aimed to establish evidence-based risk stratification, surgical indication screening and prognostic management strategies via multimodal coronary imaging evaluation in pediatric AAOCA.

Preschool children are mostly diagnosed incidentally during visits for fever, vomiting and other unrelated illnesses, whereas school-age and adolescent patients predominantly present with exertional chest tightness and pain^[8]. TTE serves as the first-line screening modality for pediatric cardiac malformations, yet its diagnostic yield for AAOCA is limited by insufficient awareness among pediatric sonographers, who prioritize cardiac enlargement and valvular lesions while neglecting detailed coronary artery screening. The initial false-negative TTE rate of 41.4% (24/58) in our cohort highlights this limitation. Typical TTE diagnostic features of AAOCA include: (1) an acute tangent angle between the anomalous coronary ostium and aortic root; (2) ectopic, superiorly positioned coronary ostia; (3) intramural course of the proximal coronary artery between the aorta and pulmonary artery; (4) linear diastolic color Doppler flow signals indicating intramural coronary segments^[9] **figure 1**.



Figure1 TTE(case 7): The right coronary artery originated from the left, adjacent to the junction of the left and right coronary valves, and originated from the left coronary sinus(white arrow).

CCTA is the gold-standard definitive imaging modality for AAOCA, providing high-resolution visualization of distal coronary vasculature and precise characterization of high-risk anatomical phenotypes(**figure2**), including interarterial/intramural coronary segment length, slit-like ostia and acute takeoff angles, with superior anatomical delineation compared with TTE^[10] (**figure3**). SPECT-MPI is subsequently performed in patients with high-risk CCTA features to assess functional myocardial ischemia. SPECT-MPI quantifies regional myocardial blood flow and cardiomyocyte viability via radiotracer distribution, enabling detection of myocardial

ischemia, injury or necrosis, and is widely used for risk stratification and prognostic assessment in adult coronary artery disease^[11]. Among the seven surgical patients in our cohort, six presented with chest pain, CCTA-confirmed ostial stenosis and intramural coronary course, alongside positive SPECT perfusion defects; six patients with normal SPECT-MPI findings were managed conservatively with serial clinical follow-up.

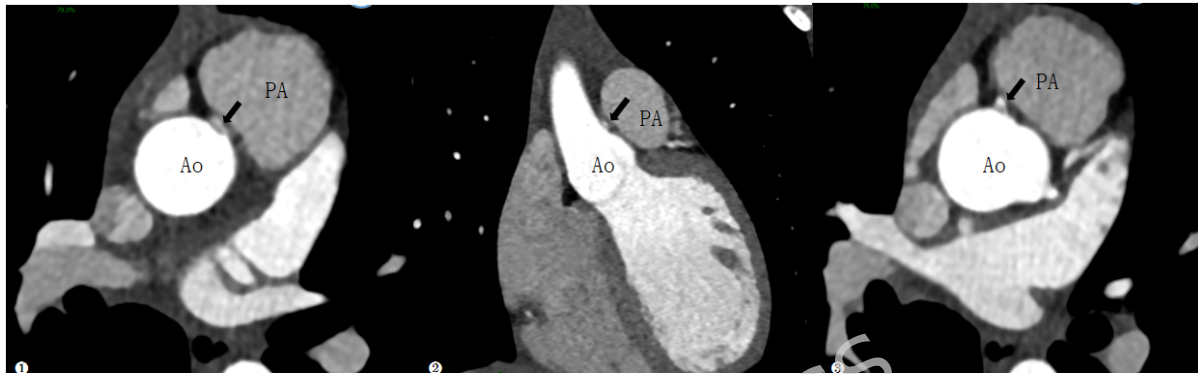


Figure2(case 7) : Intramural course AAOCA □ black arrow □: □axial CTA view shows stenosis at the origin of the right coronary artery with intramural course; □coronal CTA view demonstrates stenosis at the origin of the right coronary artery; □axial CTA view reveals that the proximal segment of the right coronary artery is wider than its origin.

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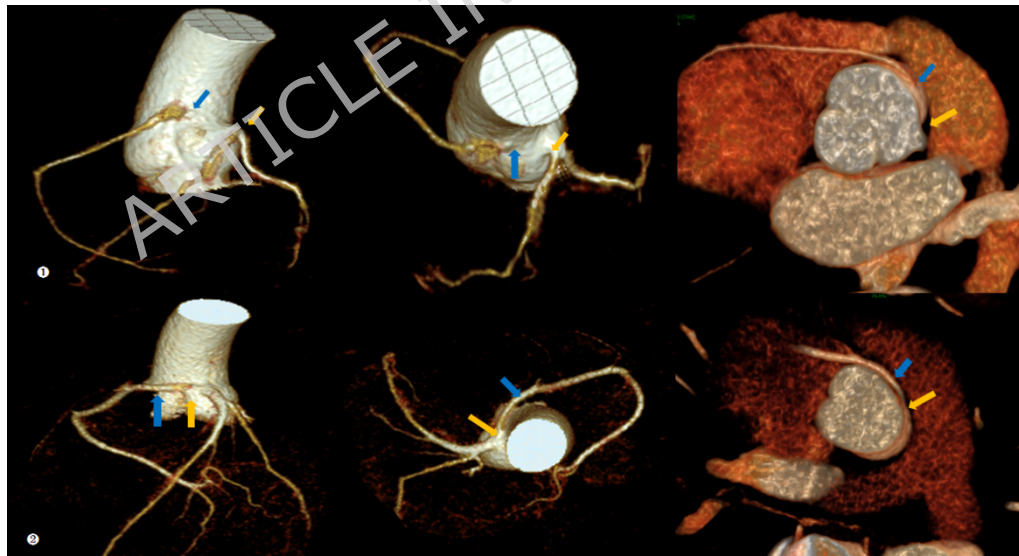


Figure3 CTA-MPVR: The yellow arrow indicates that the right coronary artery originates from the left coronary sinus, and the blue arrow shows the course of the coronary artery; □intramural: partial discontinuity can be seen(case 7); □interarterial: complete morphology can be seen

analyzed 220 children with R-AAOCA and 56 children with L-AAOCA treated at Texas Children's Hospital from 2012 to 2020 and concluded that dobutamine stress cardiac magnetic resonance (DSCMR) exhibits low positivity rate and

limited sensitivity for identifying ischemia in AAOCA patients^[10,14]. Similarly, exercise stress testing (EST) requires supplementary perfusion imaging(SPI) to capture exercise-induced ischemia in adolescents with AAOCA^[12]. A major limitation of our study is the absence of stress perfusion imaging, as caregivers declined additional testing due to concerns over radiation exposure and stress-related pharmacologic risks. Fractional flow reserve (FFR) quantifies hemodynamically significant coronary stenosis and functional impairment; Agrawal et al. reported four pediatric AAOCA patients receiving FFR assessment in 2017, yet FFR criteria are established for adult atherosclerotic coronary disease and cannot be directly extrapolated to children.

Literature focusing on pediatric SPECT-MPI for AAOCA remains scarce. Our center implemented a risk-stratified algorithm for SPECT-MPI referral: imaging is only ordered for patients with ischemia-related symptoms, abnormal ECG, elevated myocardial injury biomarkers, or high-risk CCTA anatomical features. Conversely, SPECT-MPI was not performed in patients who were completely asymptomatic and had entirely normal electrocardiography, normal laboratory test results, and no high-risk anatomical features on CTA. In these cases, the pretest probability of inducible ischemia was considered

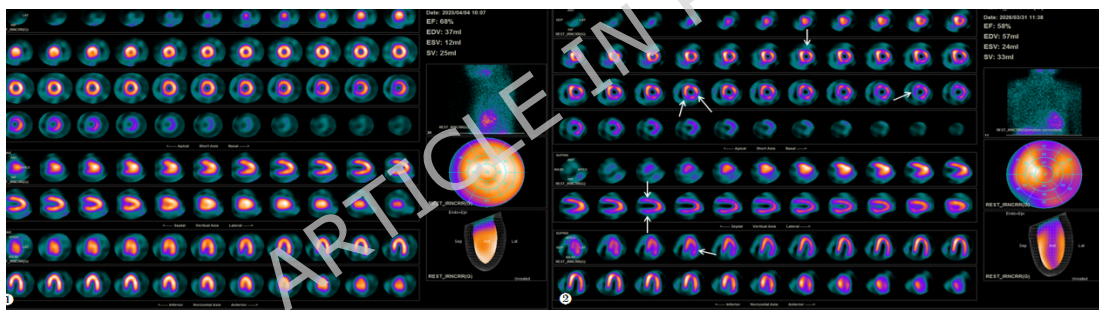


Figure 4 □ Myocardial Perfusion Scan shows normal perfusion □ □ Significant reduction of myocardial color scale distribution is seen in the anterior apical, anterior middle, inferior apical; the involved area accounts for approximately 13% of the total left ventricular area □ White arrows indicate areas of reduced uptake, suggesting surface perfusion ischemia.(case 7)

extremely low, and we deemed it unnecessary to subject these children to the radiation exposure and potential sedation risks associated with SPECT imaging. This risk-stratified approach aligns with the principle of individualized patient management and ensures that invasive or radiation-intensive investigations are reserved for those with a meaningful likelihood of ischemia. In our cohort, positive SPECT-MPI findings correlated with high-risk CCTA anatomical features, and all surgically treated patients demonstrated perfusion defects, while normal MPS results effectively ruled out the requirement for coronary surgery □ figure 4 □.

One critical limitation of standalone SPECT-MPI is the inability to precisely match ischemic myocardial segments to the culprit coronary artery. The standard AHA 17-segment myocardial model may yield false-negative results in patients with left coronary-dominant R-AAOCA, as right ventricular ischemia cannot be reliably detected. Novel three-dimensional fusion software integrating SPECT-MPI and CCTA datasets enables accurate spatial matching of perfusion defects to the corresponding anomalous coronary artery, representing a promising direction for future translational research^[13].

The 2020 European Society of Cardiology (ESC) Guidelines for the Management of Adult Congenital Heart Disease recommend surgical revascularization for all symptomatic AAOCA patients regardless of documented ischemia, and prophylactic surgery for young patients with L-AAOCA or high-risk anatomical features even in the absence of symptoms. But these recommendations are largely based on the 2008 ACC/AHA guidelines and 2017 Society of Thoracic Surgeons expert consensus statements. R-AAOCA carries a substantially lower SCD risk than L-AAOCA; surgical intervention is only indicated for L-AAOCA patients with high-risk anatomical features, while most asymptomatic R-AAOCA patients derive minimal benefit from revascularization. Therefore, surgical indications for AAOCA remain controversial worldwide.

AAOCA diagnosis often induces severe psychological anxiety in children and their guardians and impairs daily quality of life^[14]. The seven surgical patients in our cohort received revascularization after multidisciplinary discussion integrating clinical symptoms, high-risk CCTA anatomy, positive SPECT-MPI findings, international consensus guidelines and parental informed consent. We propose standardized risk communication for affected families to clarify the low absolute risk of SCD, permitting appropriate physical activity and competitive sports after comprehensive multimodal risk evaluation. Surgical correction is recommended for patients with new-onset exertional chest pain or syncope combined with high-risk coronary anatomy and positive perfusion defects on SPECT-MPI during follow-up, which balances quality of life and cardiovascular safety while alleviating caregiver anxiety.

Surgical strategies for AAOCA are tailored to individual coronary anatomy. Coronary unroofing is the primary operative technique, which reimplants the anomalous coronary ostium into the native aortic sinus with minimal coronary manipulation^[15]. However, this procedure cannot correct concomitant coronary angulation or interarterial coronary segments and carries a theoretical risk of aortic valve support structure injury and postoperative aortic regurgitation. Alternative surgical options include coronary artery reimplantation and coronary angioplasty for anatomically complex cases^[16].

4.1 Study Limitations

This study bears several inherent limitations: single-center retrospective design with a small sample size may introduce selection bias; multivariate regression analysis was not performed to adjust for confounding factors including age and comorbidities; stress perfusion imaging was not conducted for comparative functional assessment; postoperative SPECT-MPI follow-up was not completed; and the follow-up duration was insufficient to evaluate long-term ischemic outcomes. Large-sample multicenter prospective trials and randomized controlled trials are required to validate the predictive value of multimodal imaging and clarify the causal relationship between coronary anatomical phenotypes and inducible myocardial ischemia.

4.2 Future Perspectives

Collaborative multicenter prospective registries integrating comprehensive clinical profiles, exercise tolerance testing and multimodal hemodynamic imaging data are warranted to precisely delineate the correlation between coronary anatomical morphology and ischemic burden. Standardized risk stratification frameworks and personalized management algorithms will help optimize long-term cardiovascular outcomes, reduce unnecessary exercise restrictions, and avoid redundant surgical revascularization in children with AAOCA.

Acknowledgments

The authors acknowledge all clinical physicians, imaging technicians and nursing staff at Wuhan Children's Hospital for patient data collection and clinical management. We also thank all participating children and their families for their cooperation in this study.

Ethics Approval

This retrospective study was approved by the Institutional Ethics Committee of Wuhan Children's Hospital (Approval No.: 2026R050-E01). All patient data were anonymized to protect privacy. All research procedures complied with the ethical standards of the institutional review board and the 1964 Helsinki Declaration and its subsequent amendments.

Conflicts of Interest

The authors declare no conflicts of interest associated with this manuscript.

Funding

No external funding was obtained for this research.

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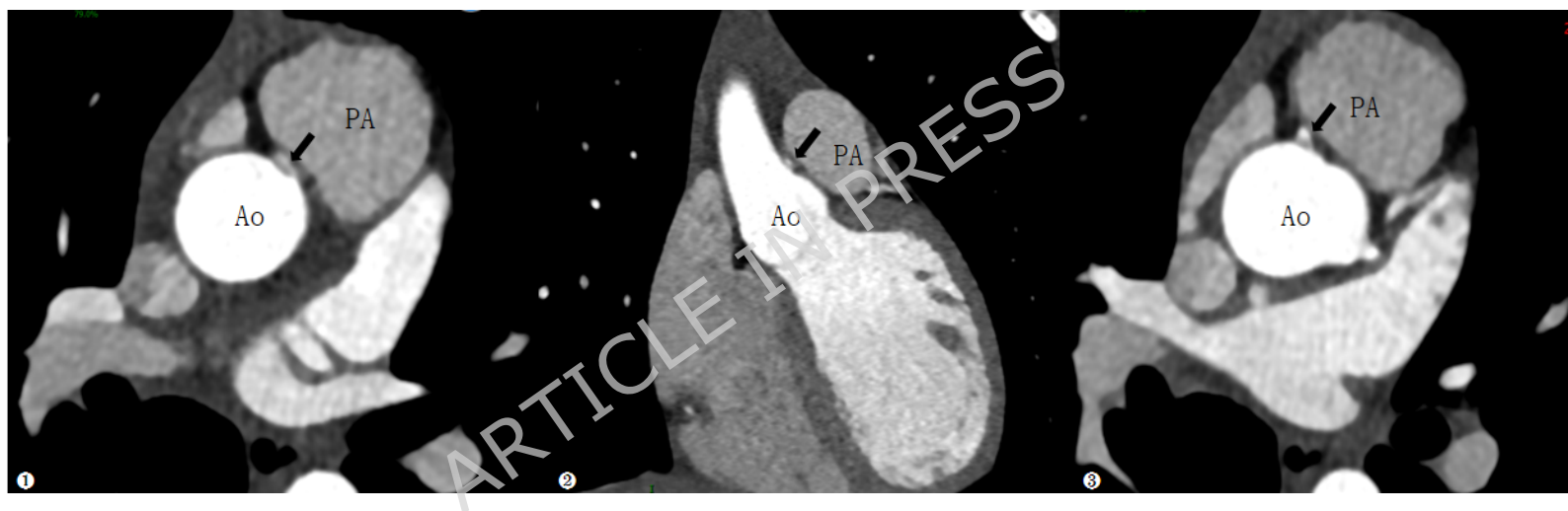
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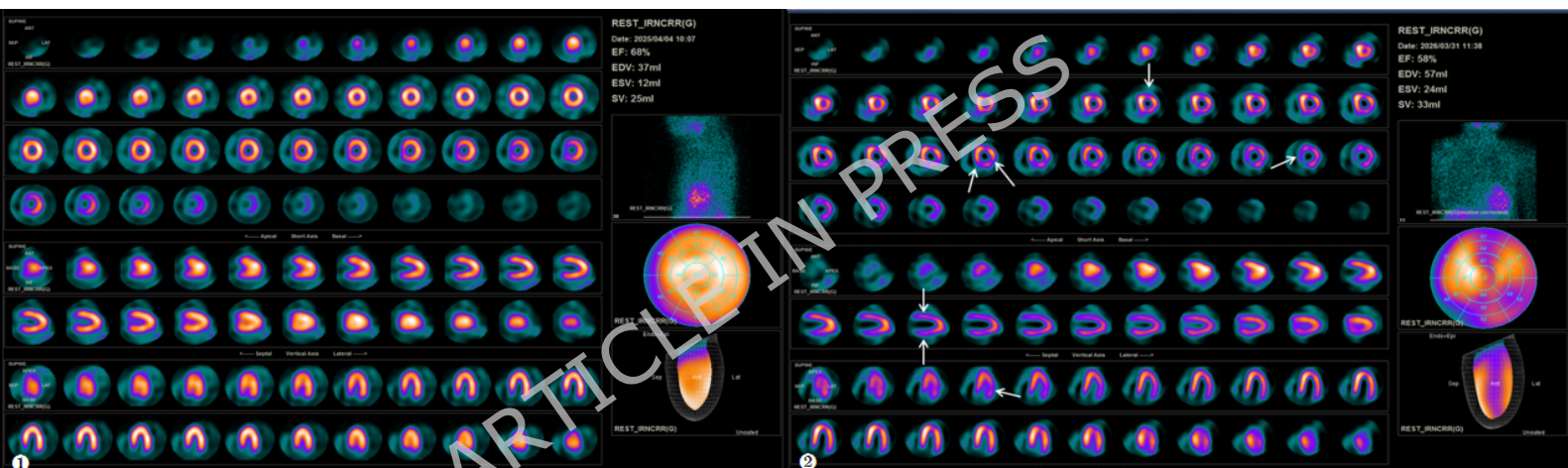


Table 1 Clinical characteristics of cases

Variable	n=58	palpitation	chest pain and tightness	syncope	related complaint
male	n=31	1 (1.72%)	16 (27.6%)	3 (5.2%)	16 (27.6%)
female	n=27	1 (1.72%)	12 (20.7%)	3 (5.2%)	6 (10.3%)
<3 years	n=0	0 (0%)	0 (0%)	0 (0%)	6 (10.3%)
3-6 years	n=0	0 (0%)	0 (0%)	0 (0%)	2 (3.4%)
6-12 years	n=0	0 (0%)	19 (32.8%)	4 (0%)	11 (19.0%)
12-18 years	n=2	2 (3.4%)	9 (15.5%)	2 (0%)	3 (5.2%)
*unrelated complaints: fever, vomiting, etc					

Table 2 Clinical characteristics of AAOCA

Variable	sort	ARCA group (n=54)	ALCA group (n=4)
sex	male	37 (68.5)	2 (50.0)
	female	17 (31.5)	2 (50.0)
clinical symptom (%)	thoracodynia	26 (48.1)	2 (50.0)
	palpitation	2 (3.7)	1 (25.0)
	syncope	5 (9.3)	1 (25.0)
	unrelated complaints*	22 (40.7)	0 (0.0)
electrocardiogram (%)	normal	36 (66.7)	2 (50.0)
	arrhythmia	13 (24.1)	1 (25.0)
	Myocardial ischemia	5 (9.3)	1 (25.0)
Cardiac CTA	Completed/positive	54 (100.0)	4 (100.0)
MPS	Completed/positive	24 (44.4)	3 (75.0)

*unrelated complaints: fever, vomiting, etc

Table 3 Clinical data of children undergo:

case	sex	Age	Action in chief	Myocardial marker	ECG	TTE
case1	Female	13 years	chest pain	normal	normal	normal
case2	Male	11 years	chest tightness	normal	normal	normal
case3	Male	6 years	Postoperative dilatation of PS	normal	Sinus arrhythmia	ALCA
case4	Female	12 years	chest tightness	normal	T-wave change	ARCA
case5	Female	10 years	chest pain	normal	normal	ARCA
case6	Male	10 year	chest pain and tightness	normal	ST segment elevation	Right crown high origin
case7 (Figure 1)	Male	9 years	chest tightness and pain	normal	Ectopic room rule, no ST change	normal

ECG, electrocardiogram; TTE, echocardiography; CTA, coronary computed tomography angiogram

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	MPS (reduction area of myocardial color scale CTA distributi on, and percent of the total left ventricula r area)	treatment
ALCA, interarter ial	anterior wall near the apex of the left ventricle, 4%	angioplast y
ARCA, proximal narrowing	apex of the heart, anterior parapical segment, middle part, 14%	unroofing
ALCA, proximal narrowing □ interarter	apex of the heart, 2%	unroofing
ARCA, intermural □ proximal narrowing	in the apex of the heart, anterior wall and the middle part of the heart, 4%	unroofing
ARCA, intermural □ proximal narrowing	in the anterior apical segment and the middle part of the left ventricle □ 13%	unroofing
ARCA, intermural □ proximal narrowing	in the proximal apical segment of the lower wall of the left ventricle, 1%	unroofing
ARCA, intermural □ proximal narrowing	in the anterior apical, anterior middle, inferior apical ; 13%	unroofing

phy; MPS, single photon emission tomog